

EE 660 Power Management IC Design

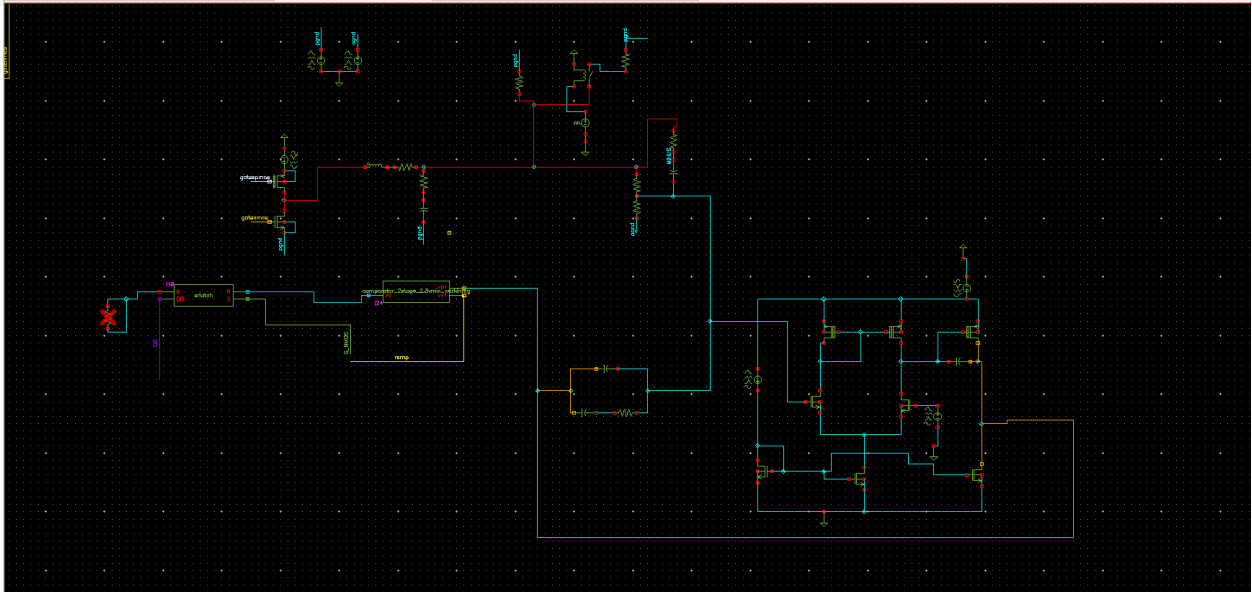
Design of a closed-loop voltage-mode buck converter

SAHASRITH BOOTLA (23110064)

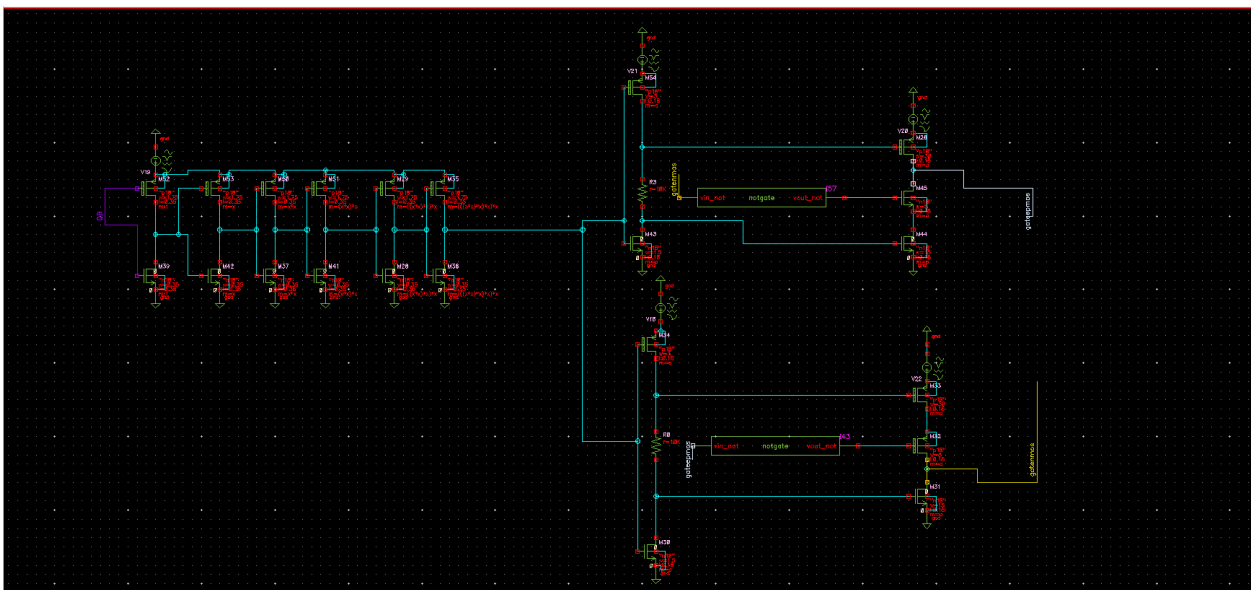
Part 3 : Integration and optimization

Integrate all elements together to complete your design and tune the parameters to meet the specifications!

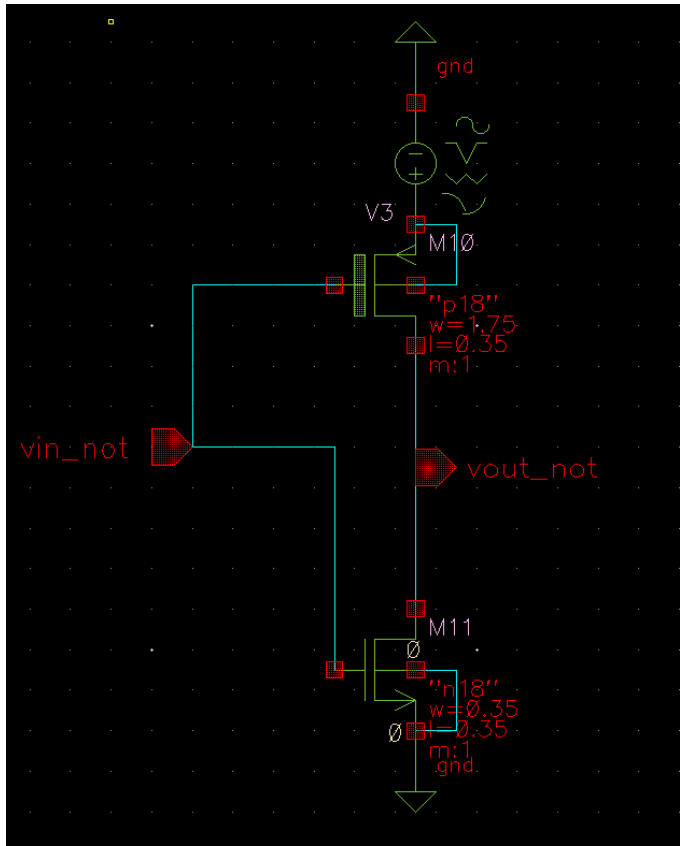
CLOSED LOOP SCHEMATIC :



DRIVER CIRCUIT :

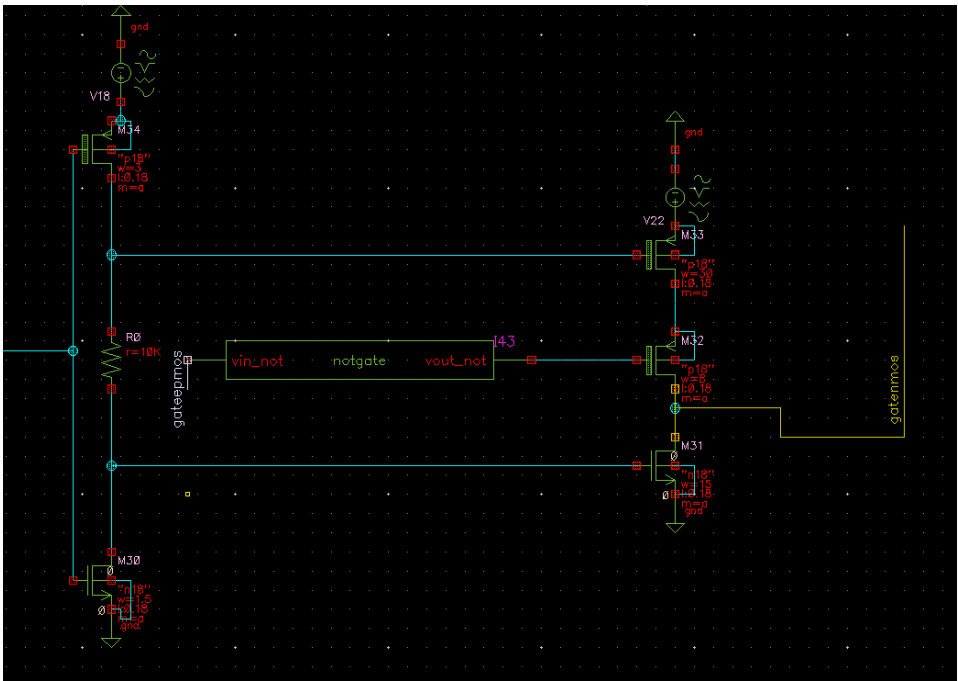


NOT GATE used in driver ckt :



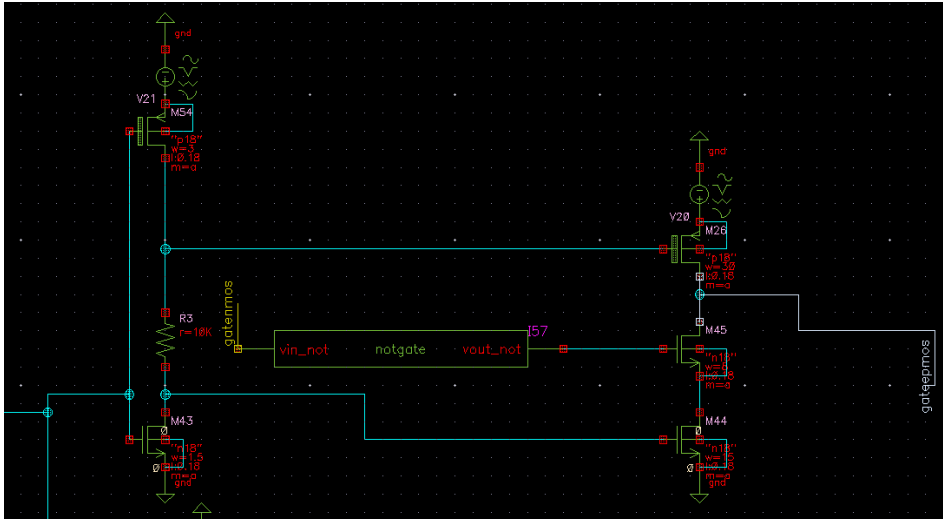
MOSFET	Type	W (μm)	L (μm)
M10	PMOS	1.75	0.35
M11	NMOS	0.35	0.35

Gate nmos :



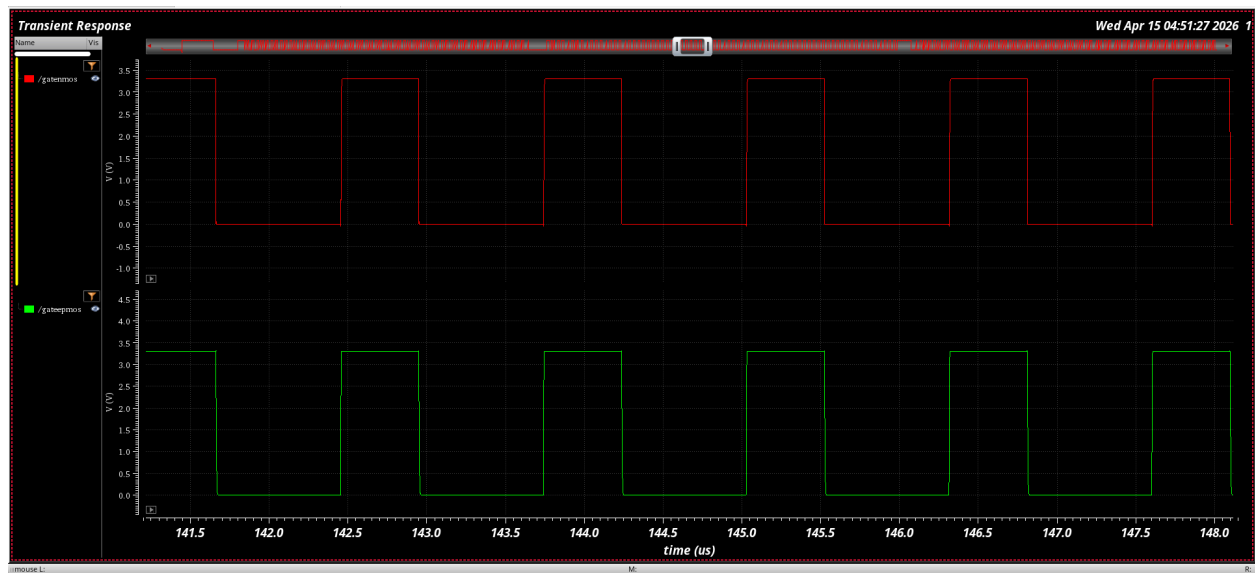
MOSFETs	Type	W (μm)	L (μm)
M34	PMOS	3	0.18
M33	PMOS	30	0.18
M31	NMOS	15	0.18
M30	NMOS	1.5	0.18
M32	PMOS	8	0.18

Gate pmos :

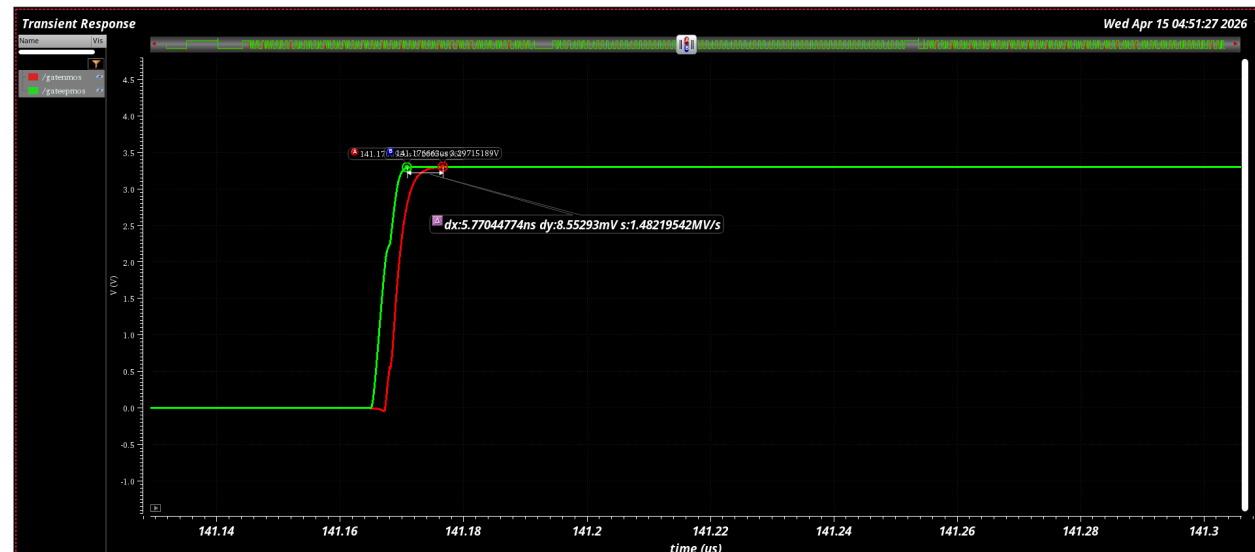


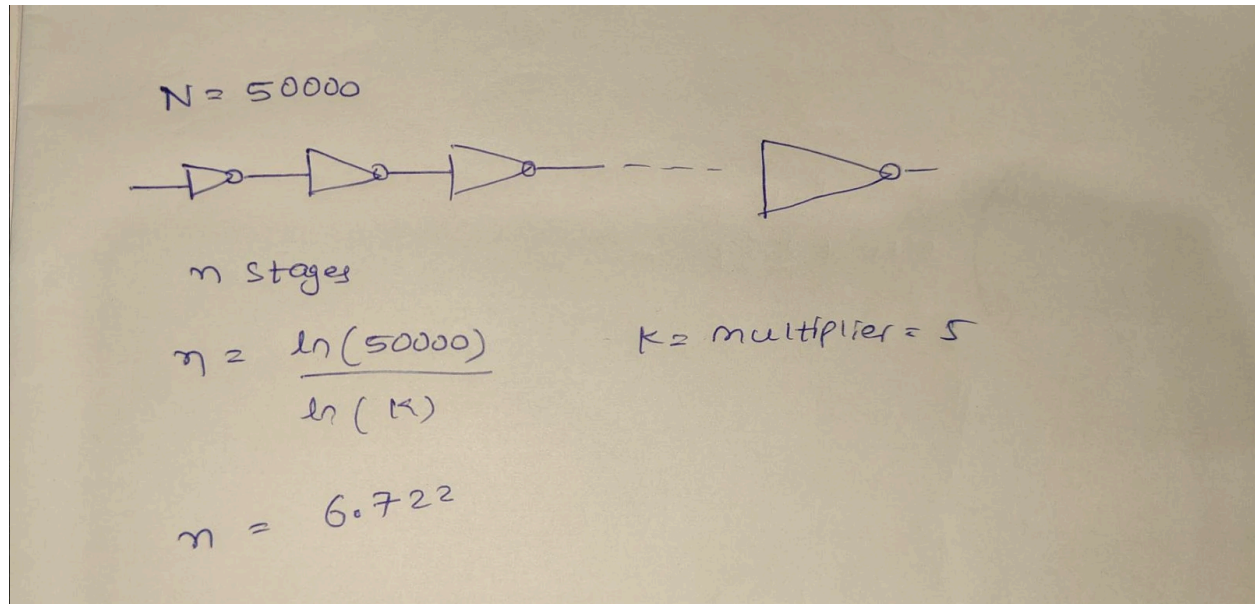
MOSFETs	Type	W (μm)	L (μm)
M54	PMOS	3	0.18
M26	PMOS	30	0.18
M44	NMOS	15	0.18
M43	NMOS	1.5	0.18
M45	NMOS	8	0.18

OUTPUT :



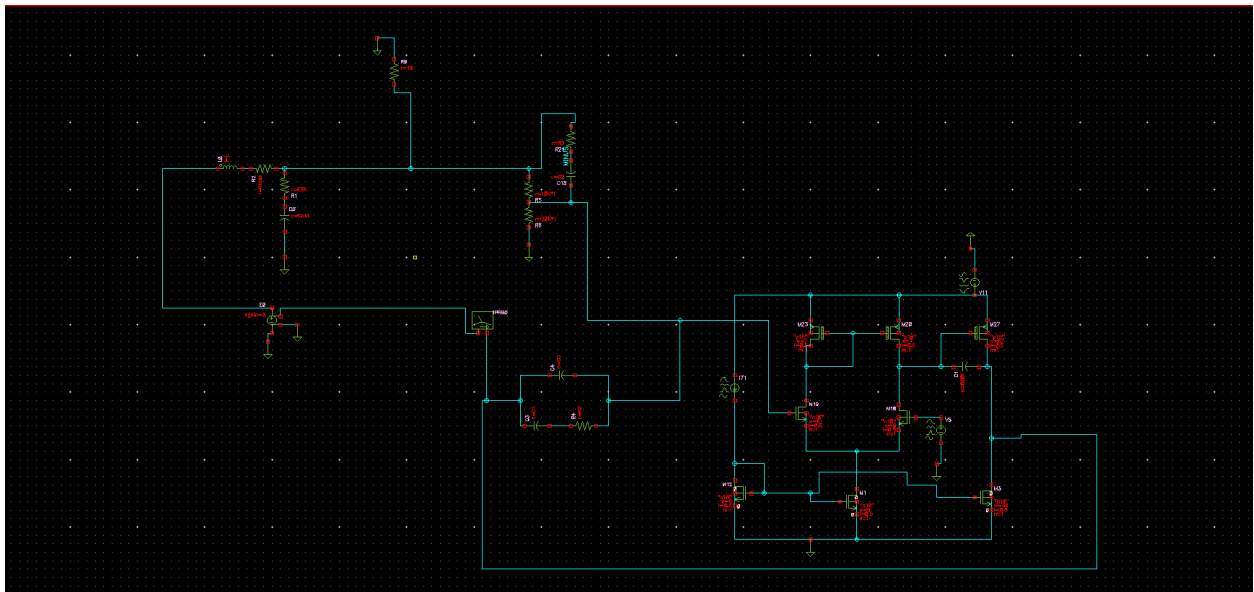
Deadtime = 5.7ns

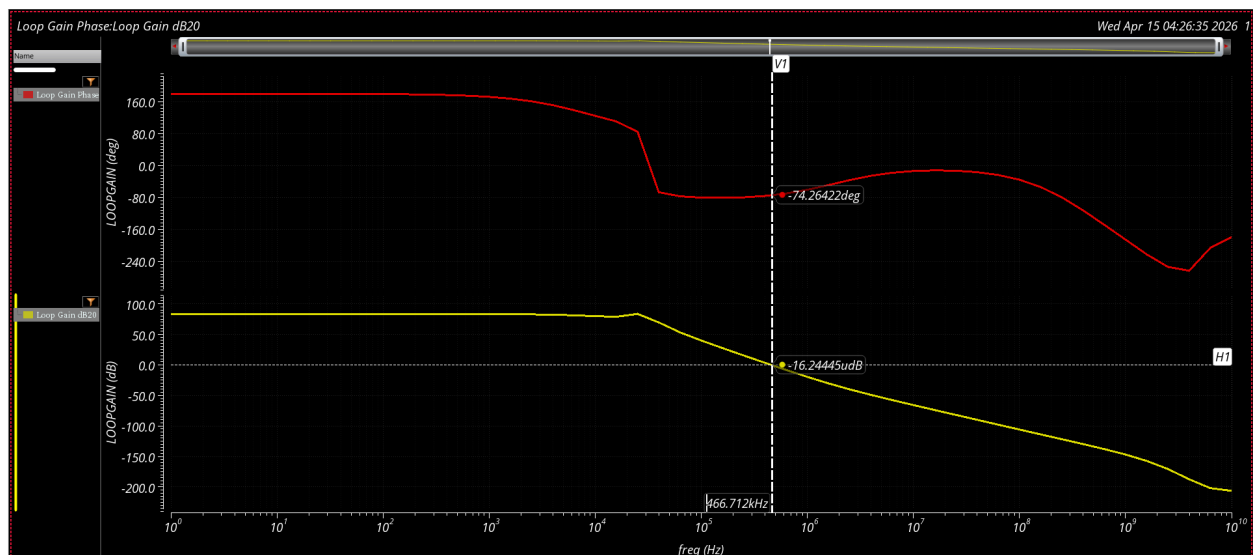




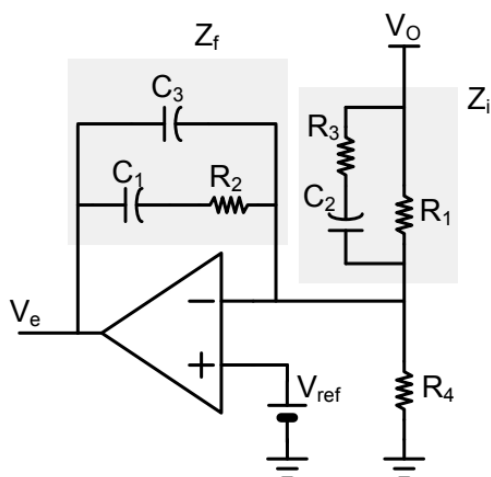
STB ANALYSIS

1. Before compensation

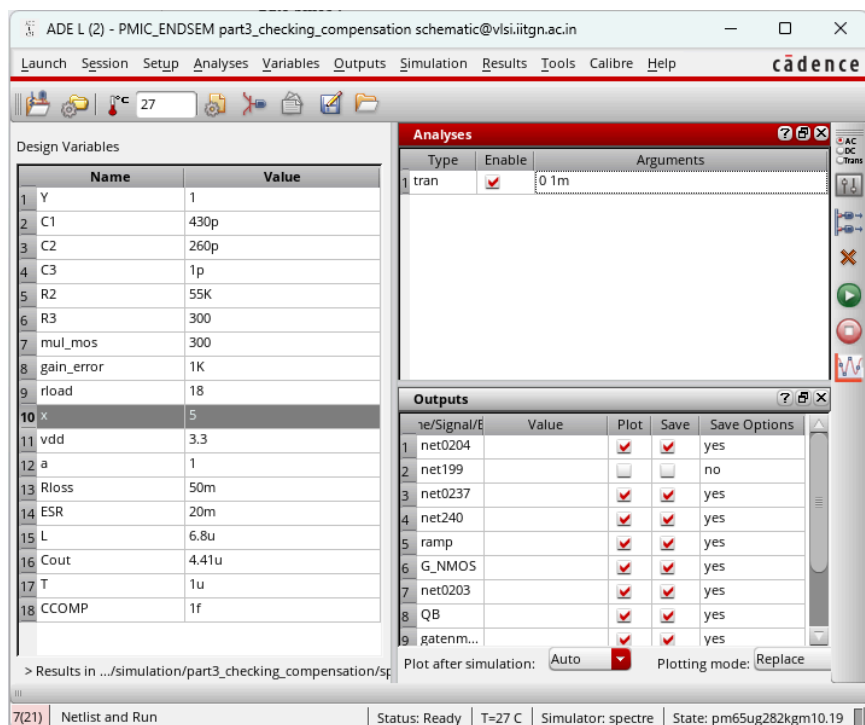


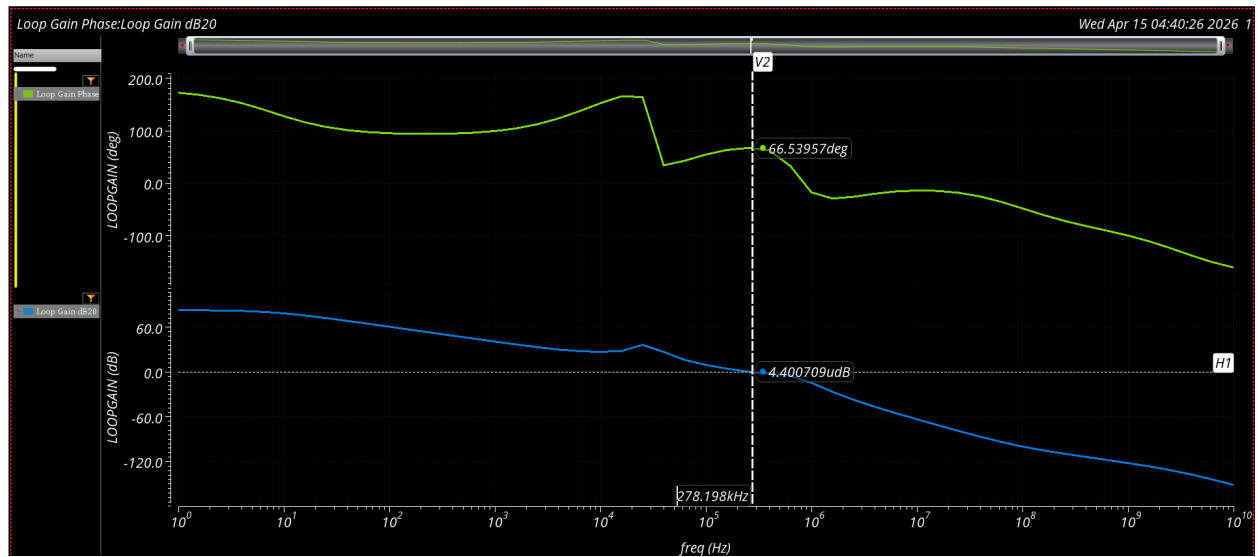


2. After TYPE-III compensation



Component	Value
R1	10 kΩ
R4	20 kΩ
R2	55 kΩ
R3	300 Ω
C1	430 pF
C2	260 pF
C3	1 pF

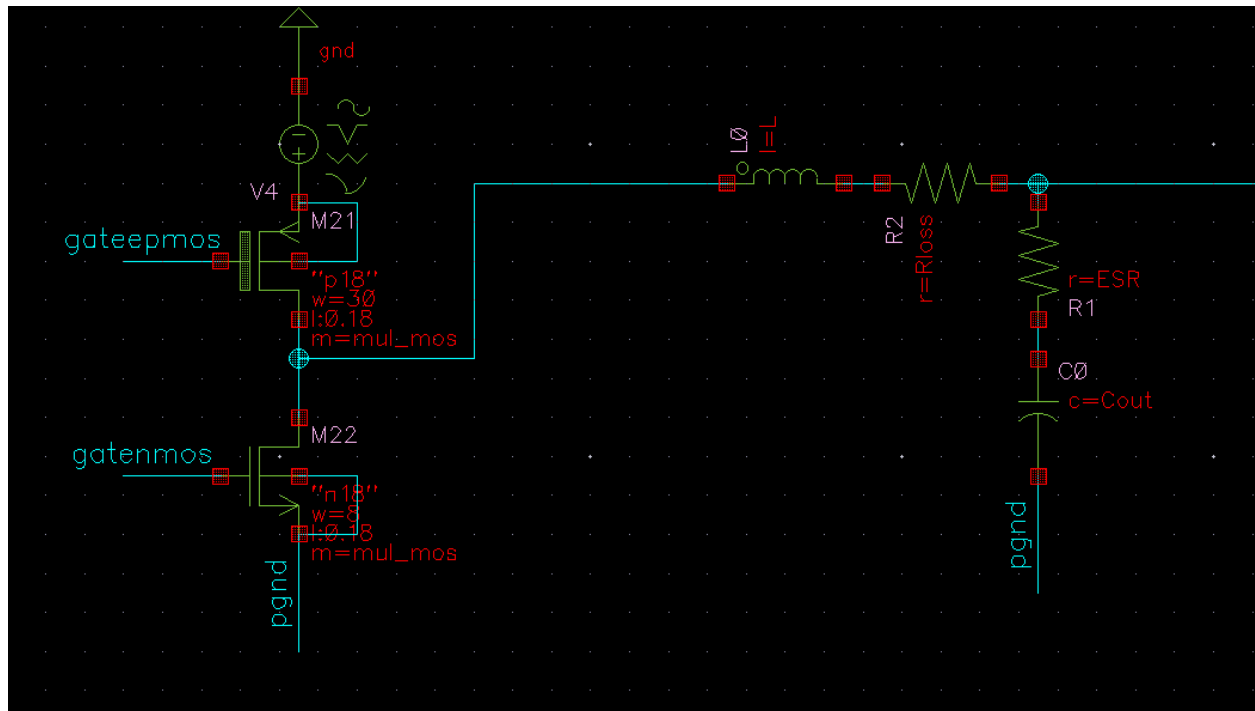




```

stb: freq = 10 GHz      (100 %), step = 3.69 GHz      (2 %)
Gain margin = 10.6427 dB at frequency = 873.609 kHz.
Phase margin = 66.247 Deg at frequency = 268.827 kHz.
Accumulated DC solution time = 0 s.
Intrinsic stb analysis time = 20 ms.
Total time required for stb analysis 'stb': CPU = 8.479 ms, elapsed = 43.3168 ms.
Time accumulated: CPU = 261.165 ms, elapsed = 1.32483 s.
Peak resident memory used = 68.8 Mbytes.
  
```

POWER MOSFETS W/L DESIGN :



$$\text{PMOS} \Rightarrow \frac{W}{L} = \frac{30 \times 300}{0.18} = \frac{9000}{0.18} = 50000 \text{ (mul_mos} = 300)$$

$$\text{NMOS} \Rightarrow \frac{W}{L} = \frac{8 \times 300}{0.18} = \frac{2400}{0.18} = 13333$$

we should calculate w, L of power MOSFets from efficiency $\eta = \frac{P_{out}}{P_{in}}$

$$\eta > 92\% \Rightarrow \eta = \frac{P_{out}}{P_{out} + P_{loss}}$$

$$\left(\frac{1}{\eta}\right) P_{loss} = \left(\frac{1}{\eta} - 1\right) P_{out}$$

$$P_{loss} = \left(\frac{1}{0.92} - 1\right) P_{out}$$

$$P_{loss} = (0.0869) P_{out}$$

$$P_{in} = (3.3) (800 \times 10^{-3}) = 2.64 \text{ W}$$

$$P_{loss} = (0.0869) (2.64) = 229.4 \text{ mW}$$

$$P_{conduction} \geq 90\% (P_{total})$$

$$\Rightarrow (0.9) (229.4)$$

$$\Rightarrow 206.46 \text{ mW} = I_{Load}^2 R_{loss}$$

$$R_{loss} = \frac{206.46}{(0.8)^2} = 322.59 \text{ m}\Omega$$

$$R_{loss} = R_{esr} + R_{on} = 50 \times 10^{-3} + R_{on}$$

$$R_{on} = 272.6 \text{ m}\Omega$$

$$I_{ds} = \mu_n C_{ox} \left(\frac{W}{L} \right) \left[(V_{gs} - V_{th}) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

Our MOSFET is in linear region

$$\mu_n C_{ox} \left(\frac{W}{L} \right) (V_{gs} - V_{th}) V_{ds}$$

$$R_{ds} = \frac{V_{ds}}{I_{ds}} = \frac{1}{\mu_n C_{ox} \left(\frac{W}{L} \right) (V_{gs} - V_{th})}$$

$$\text{PMOS} \Rightarrow \frac{1}{(65 \mu) \left(\frac{W}{L} \right) (1.5 - 0.45)} = 273 \text{ m}\Omega$$

$$V_{gs} = |1.8 - 3.3| = 1.5$$

$$\frac{W}{L} = \frac{1}{(65 \times 10^{-6}) (273 \times 10^{-3})}$$

$$\frac{W}{L} = \frac{1}{(65 \times 10^{-6}) (273 \times 10^{-3}) (0.55)}$$

$$\boxed{\frac{W}{L}_{\text{PMOS}} = 36357}$$

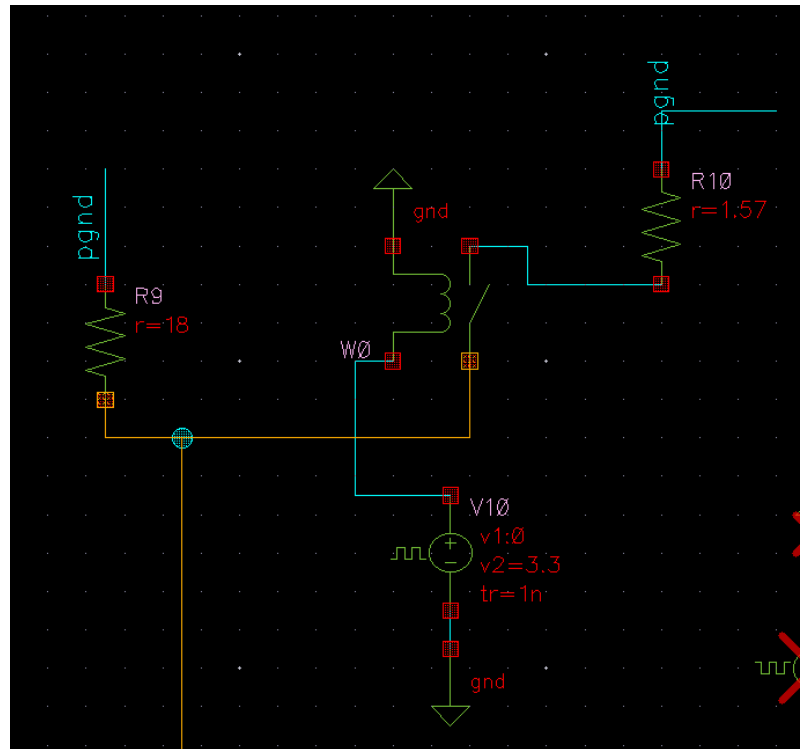
$$\text{NMOS} \Rightarrow \frac{W}{L}_{\text{NMOS}} = \frac{1}{(300 \times 10^{-6}) (1.8 - 0.45) (273 \times 10^{-3})}$$

$$\boxed{\left(\frac{W}{L} \right)_{\text{NMOS}} = 9044}$$

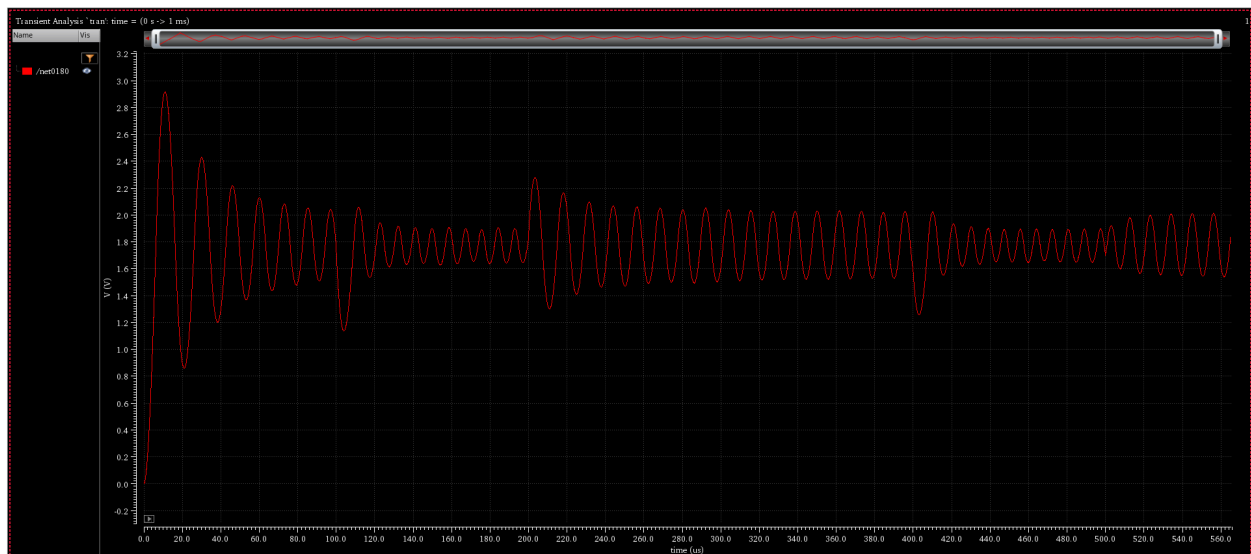
LOAD TRANSIENT :

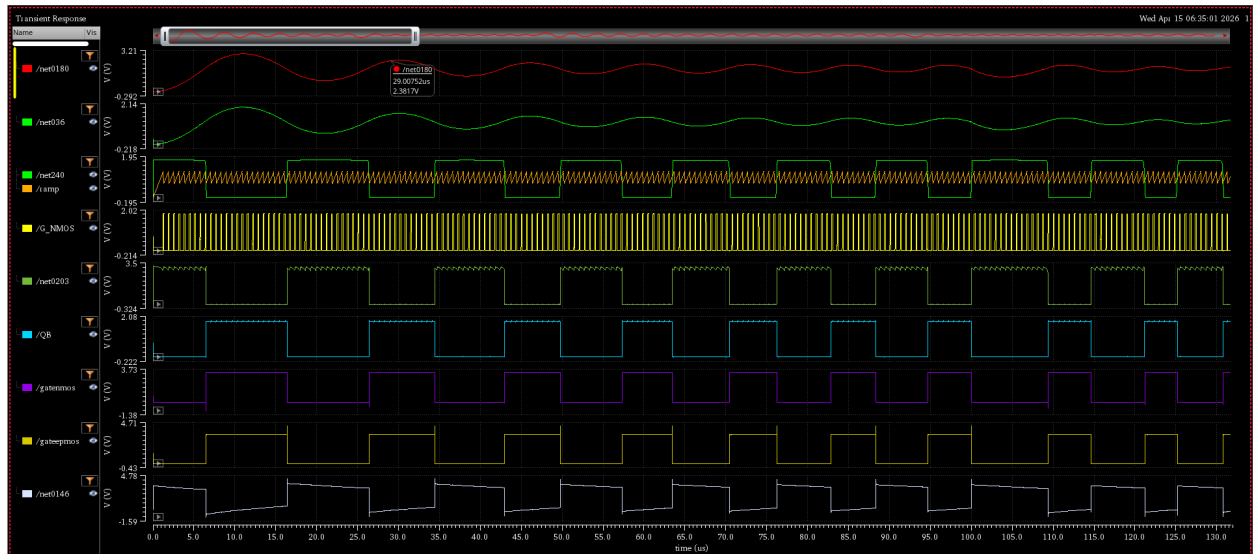
1. Load block

For every 100 us the load
Will change from 100mA
to 800mA

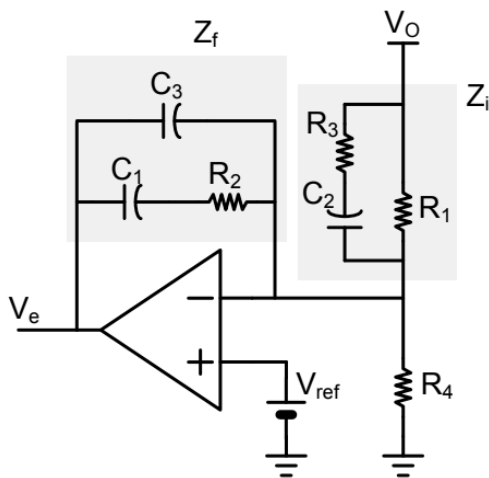


2. Regulated output before compensation:

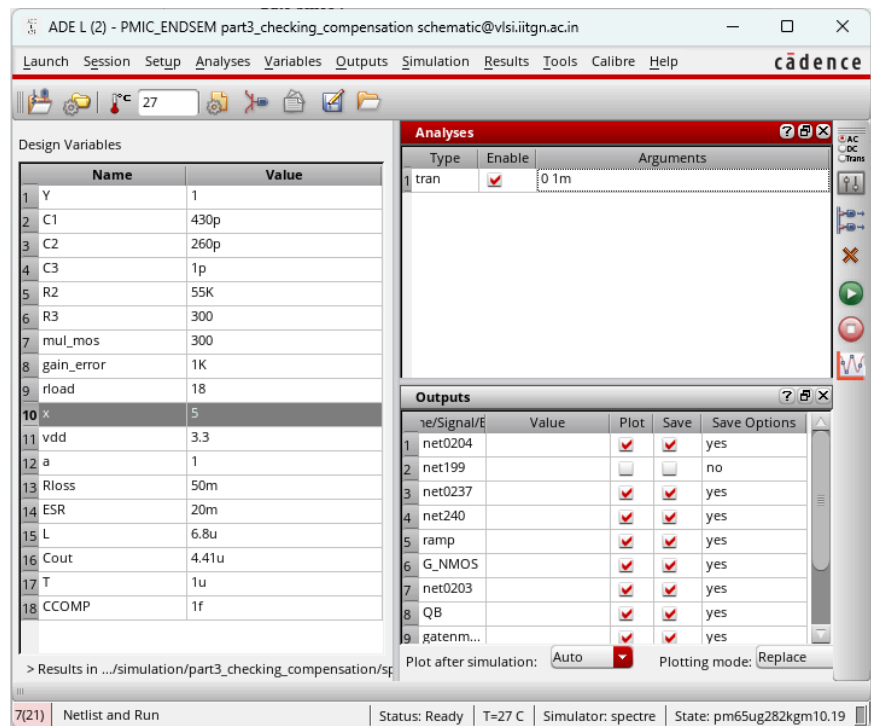




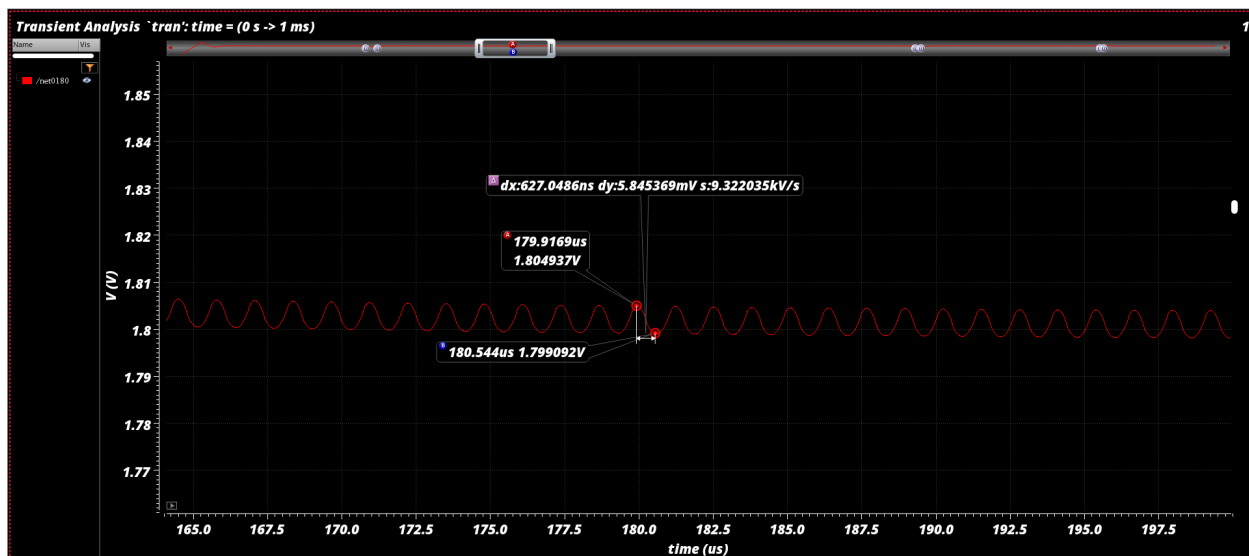
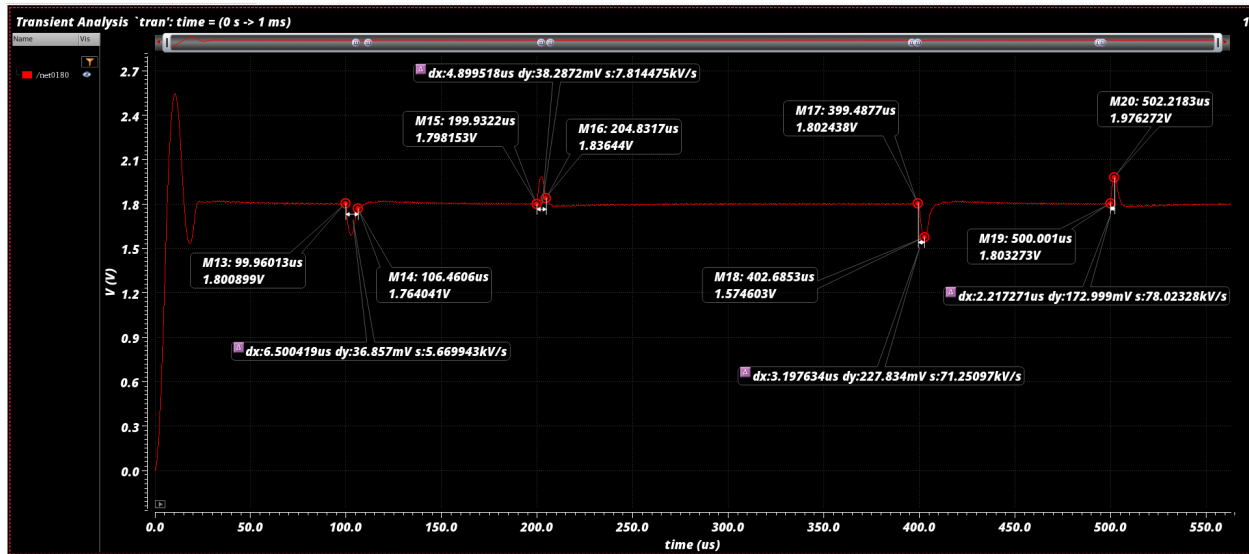
3. After TYPE-III compensation



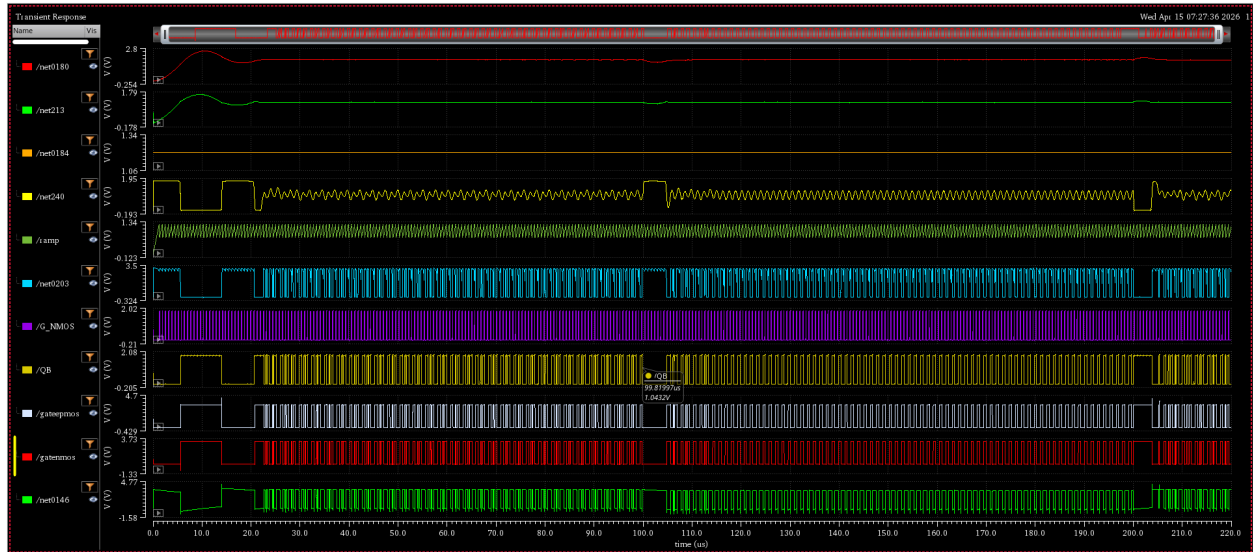
Component	Value
R1	10 k Ω
R4	20 k Ω
R2	55 k Ω
R3	300 Ω
C1	430 pF
C2	260 pF
C3	1 pF



3. Regulated output after compensation:



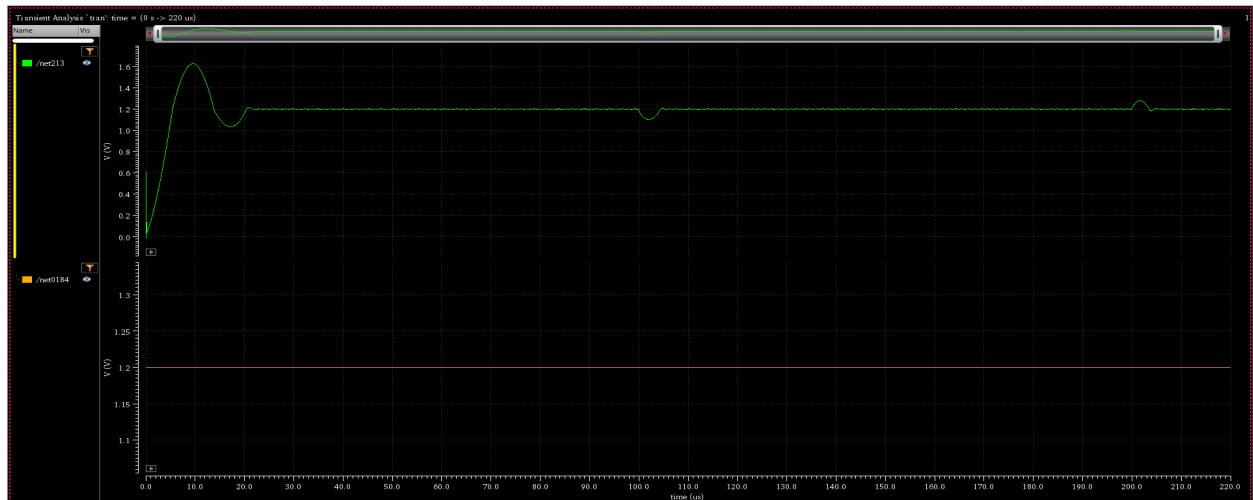
Parameter	Value (Approx)	Notes
Overshoot (up)	172.9 mV	-
Undershoot (down)	227.8 mV	-
Settling time 1	4.89 μ s	From zoomed waveform (oscillations damp out)
Settling time 2	6.5 μ s	From zoomed waveform (oscillations damp out)
Steady-state ripple	5.8 mV	From second image



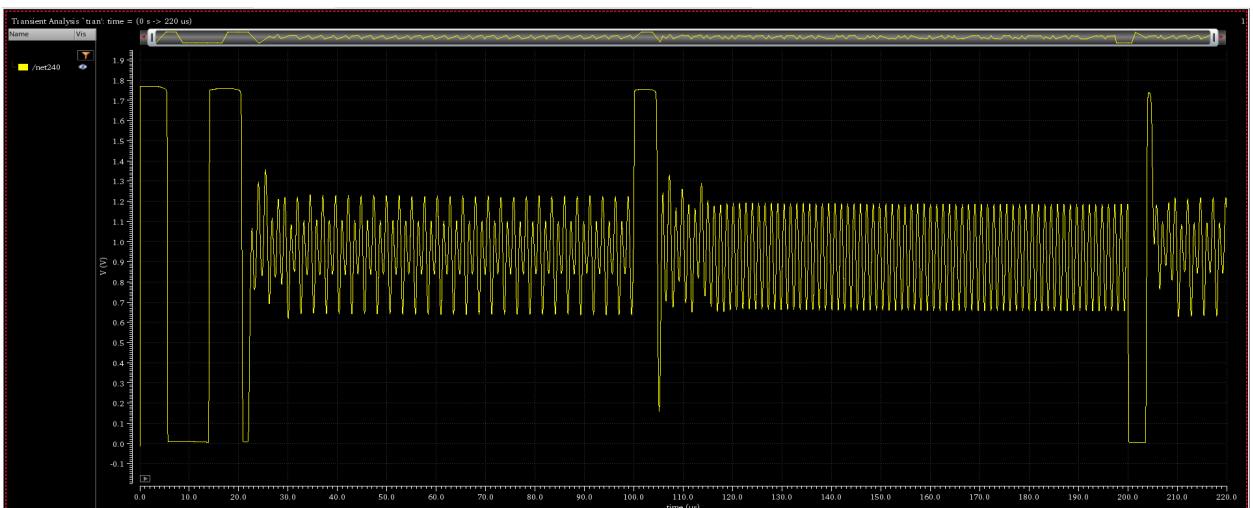
Now we will check the inputs and outputs of each block in the compensated voltage mode SWR

1. Error amplifier (2 stage OTA):

Input:

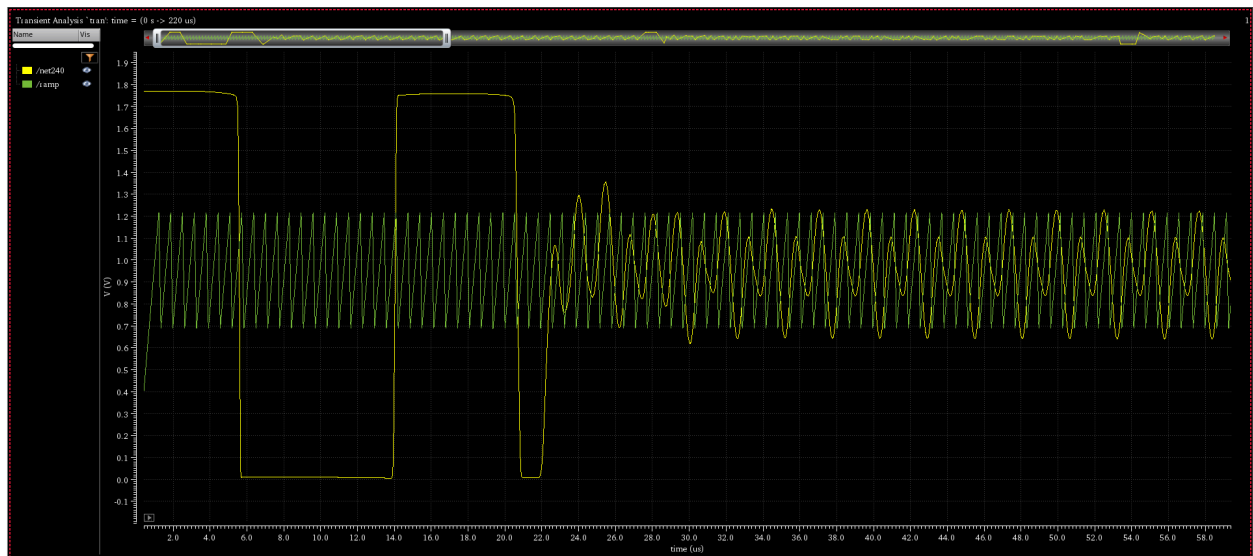
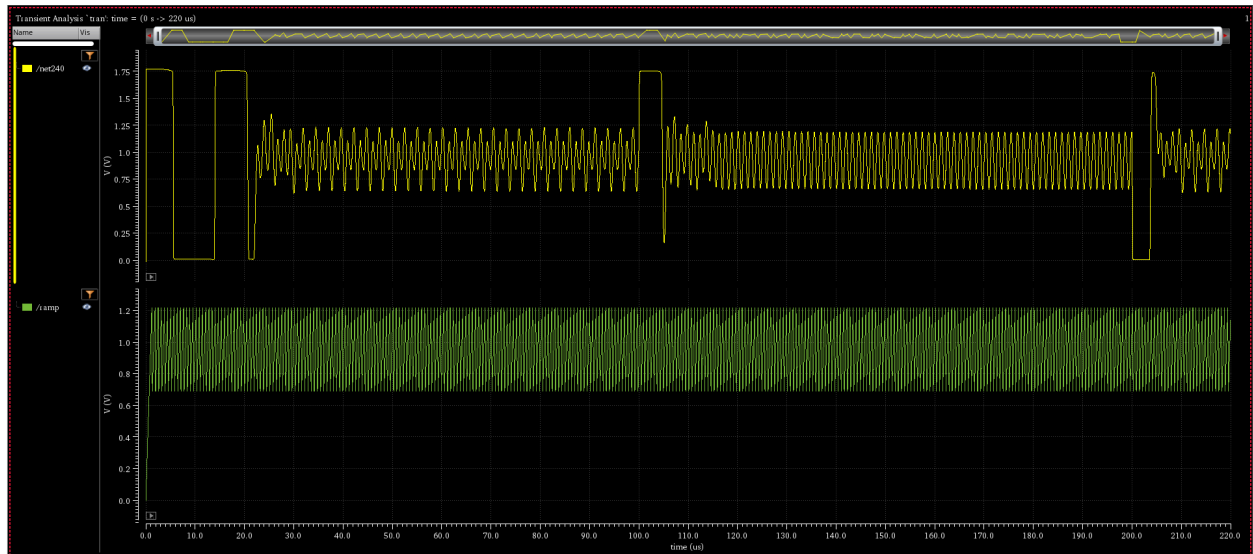


Output:

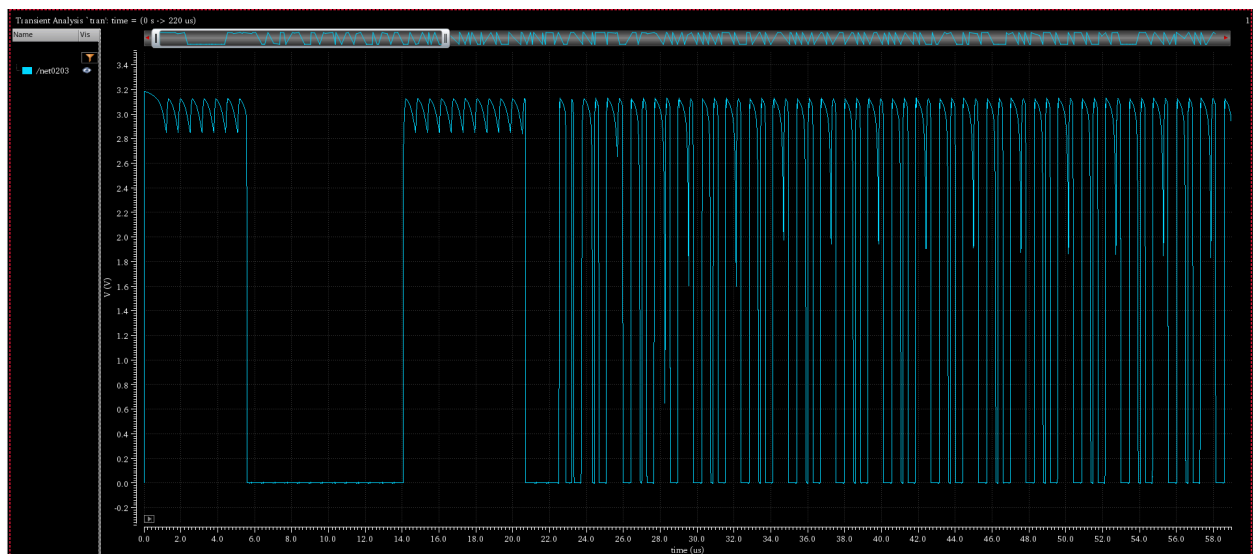


2. Comparator:

Input:

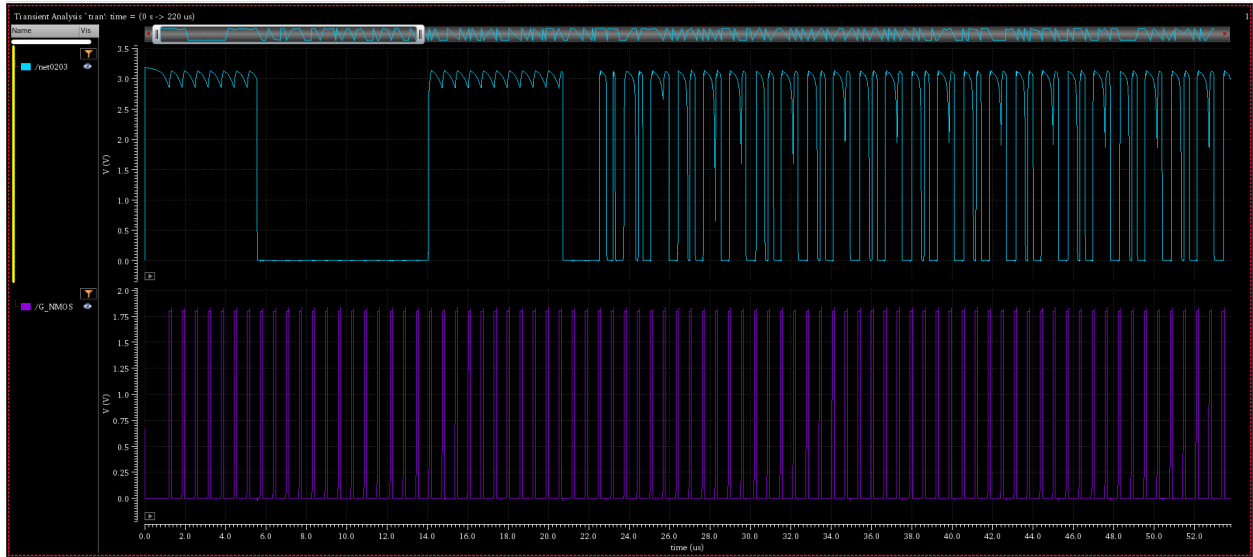


Output:

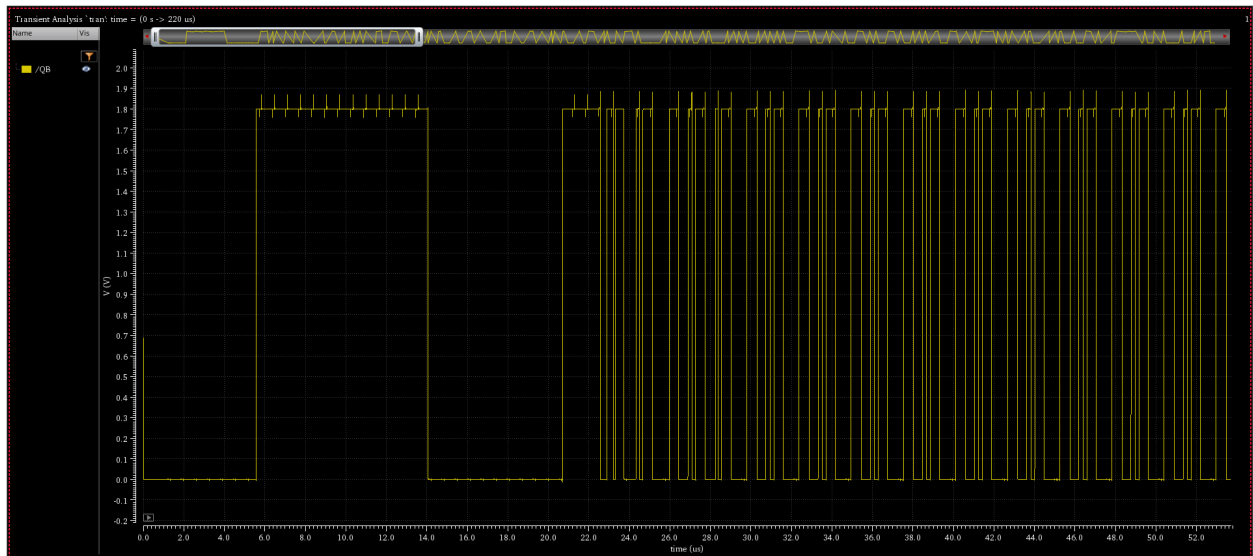


3. SR LATCH

Input:

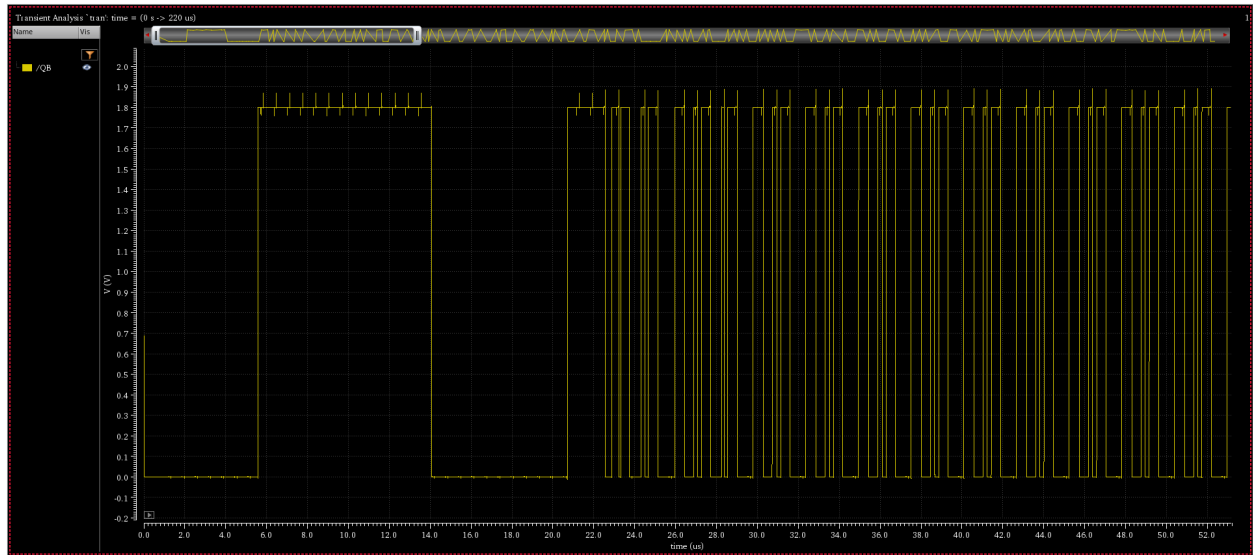


Output:

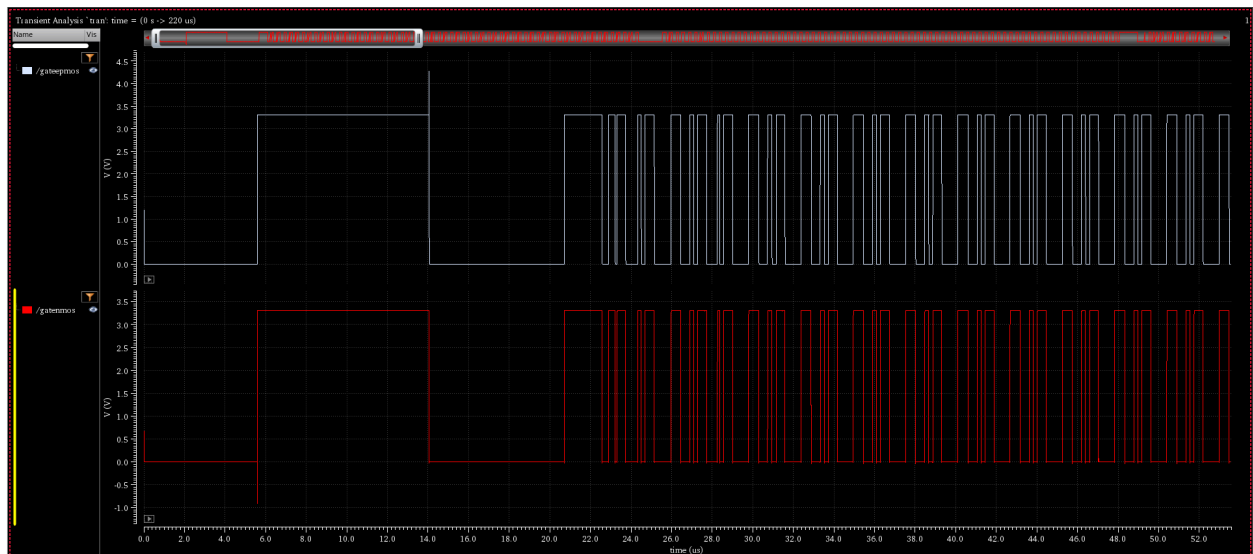


4. GATE DRIVER

Input:

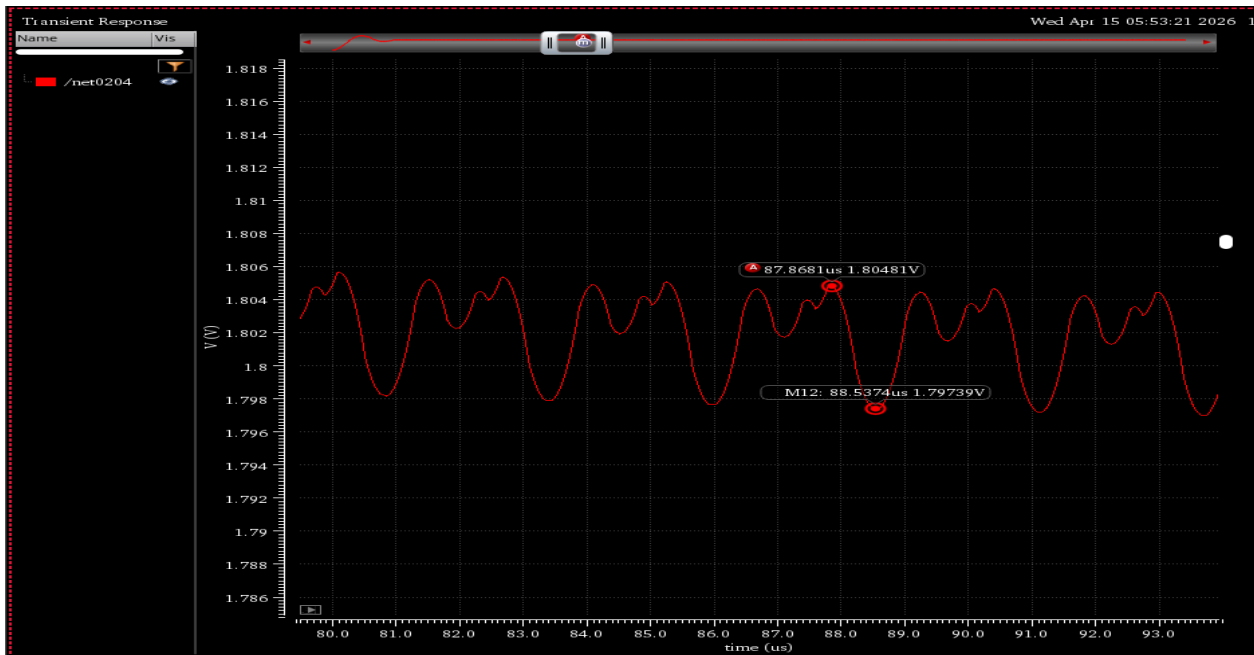
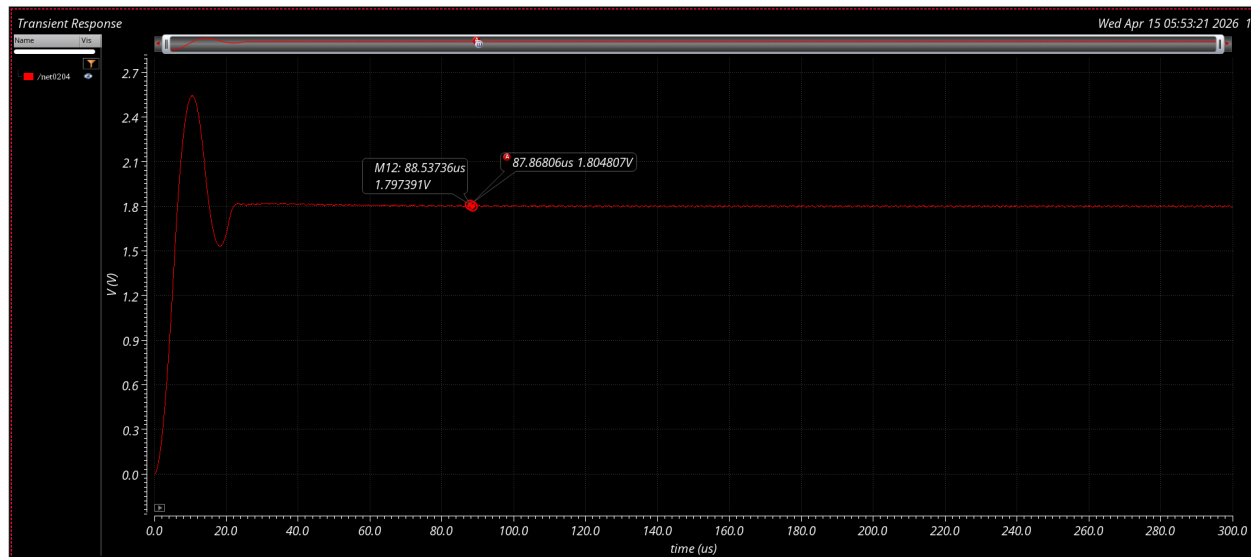


Output:



DC STEADY STATE VOLTAGE ERROR :

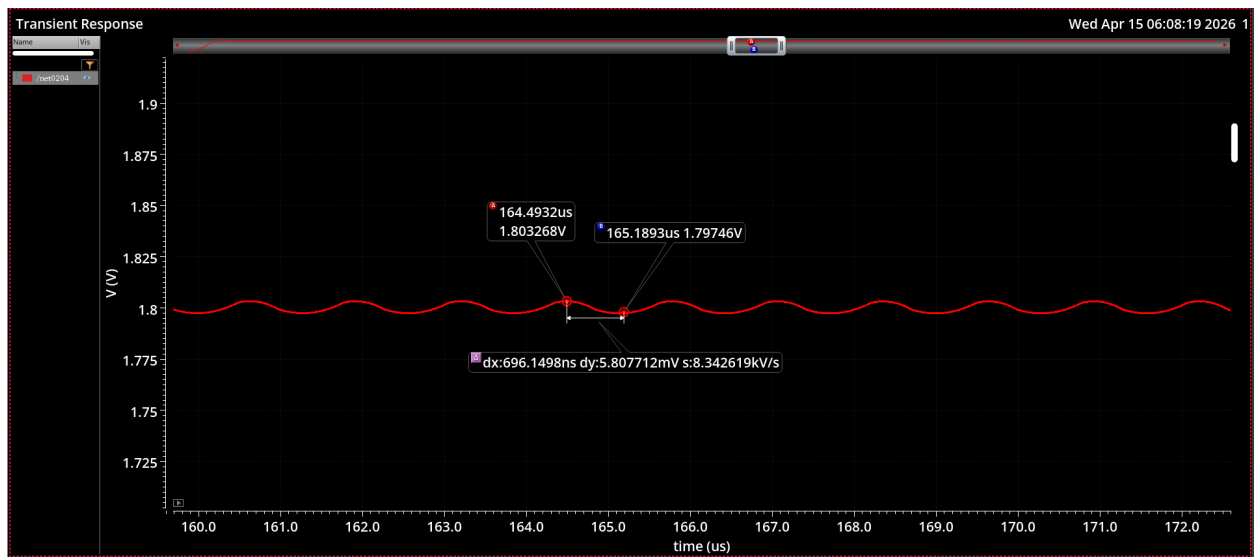
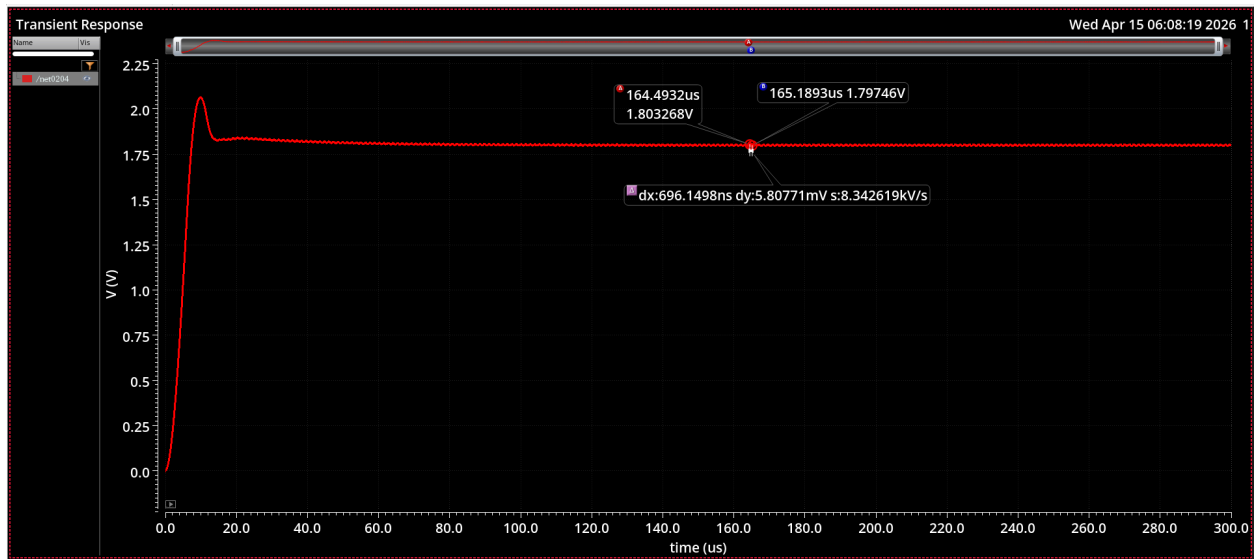
1. For light load (100mA) :



$$\text{Steady State Error min} = 1.8 - 1.79739 = 0.00261\text{V}$$

$$\text{Steady State Error max} = 1.804807 - 1.8 = 0.004807\text{V}$$

2. For Heavy load (800mA) :

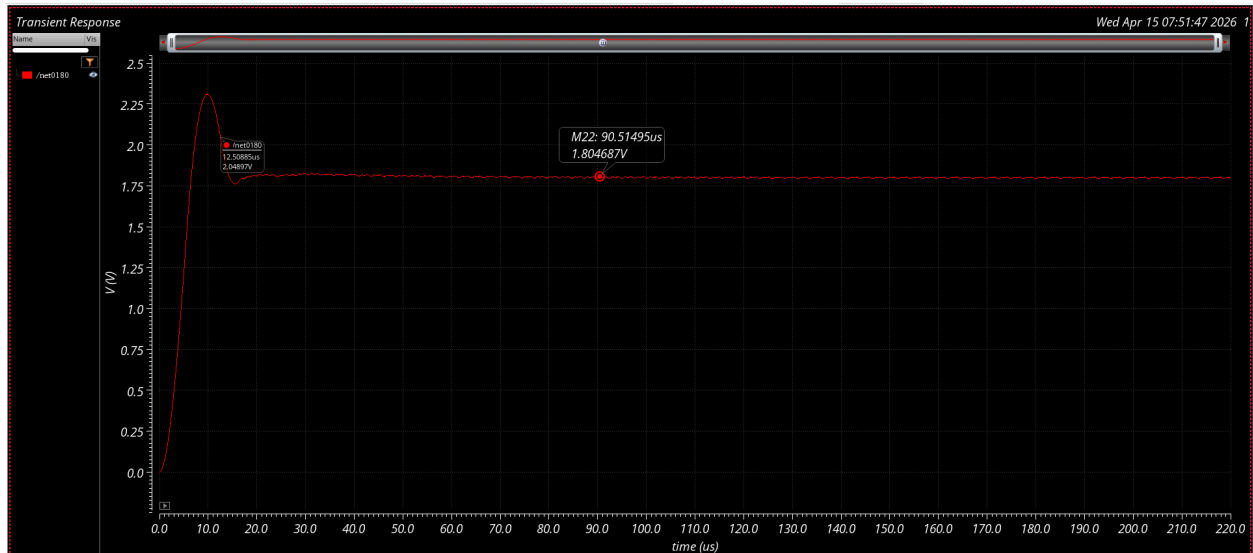


$$\text{Steady State Error min} = 1.8 - 1.79746 = 0.00254\text{V} < 0.2 \% (1.8)$$

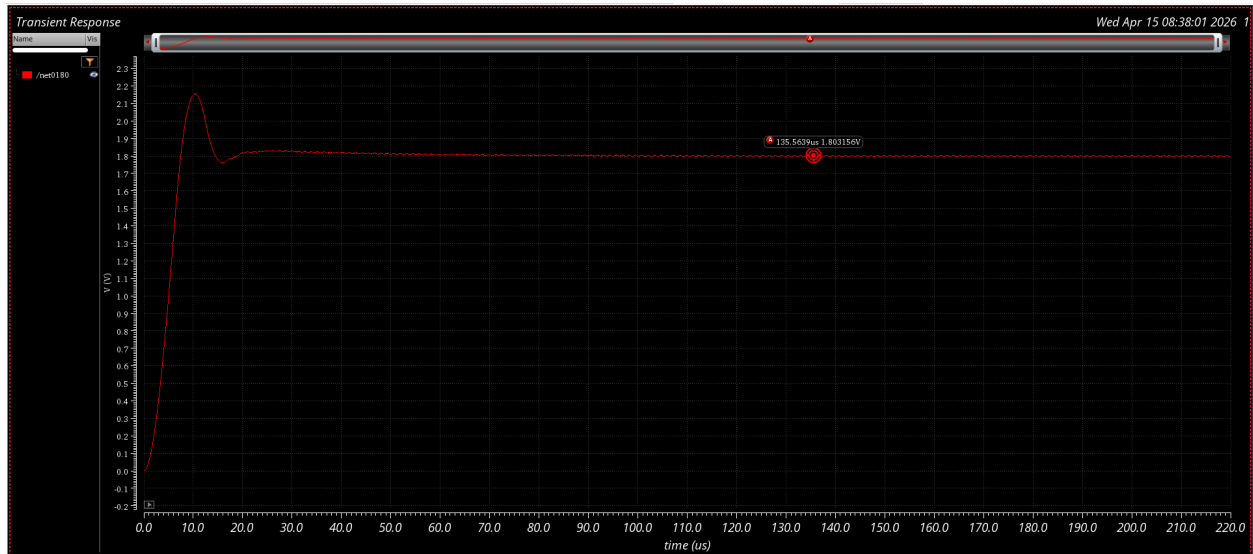
$$\text{Steady State Error max} = 1.803268 - 1.8 = 0.003268\text{V} < 0.2 \% (1.8)$$

LINE REGULATION (For nominal load) :

1. Input Voltage = 3.63V



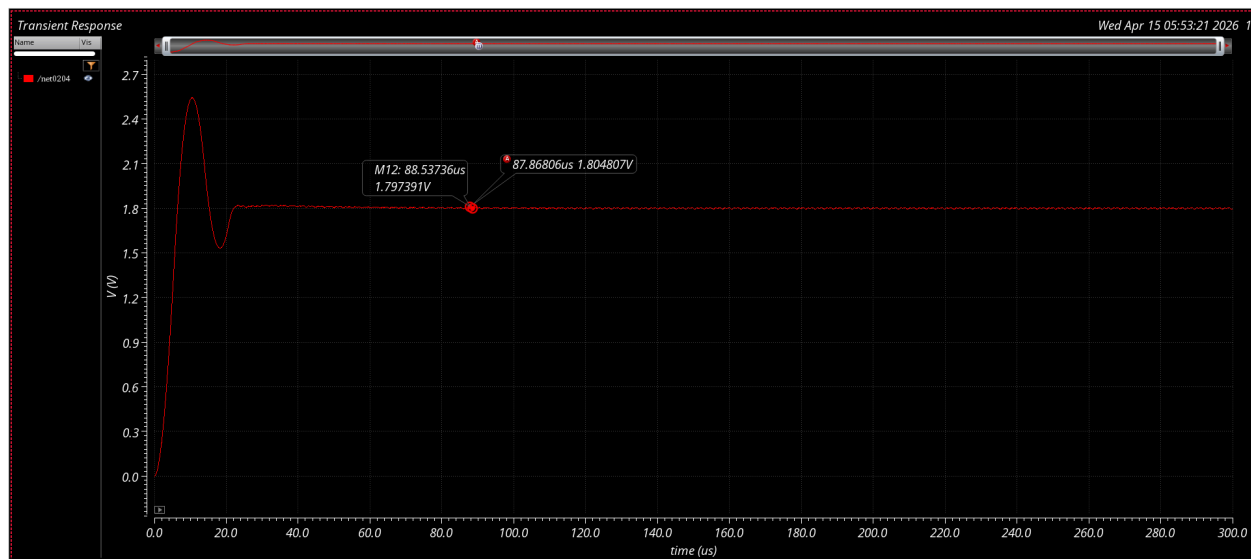
2. Input Voltage = 2.97V



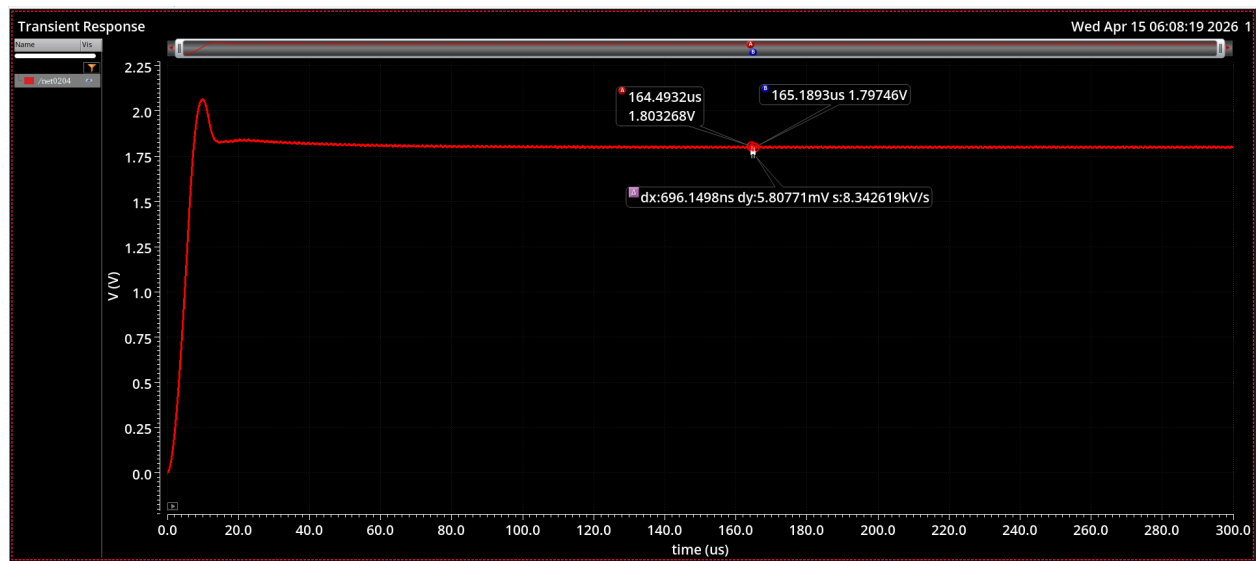
$$\text{Difference} = 1.804687 - 1.80315 = 0.001537 \leq 5\% (1.8)$$

LOAD REGULATION :

For light load (100mA) :



For Heavy load (800mA) :



$$\text{Difference} = 1.804807 - 1.803268 = 0.001539 \leq 5\% (1.8)$$